

SAMEER A DAMBAL

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Research Overview

1. Designing noise-assisted quantum algorithms.
2. Developing theoretical techniques to harness photon entanglement in quantum spectroscopy.

Education

University of Houston

August 2022 – Present

Ph.D. in Physics advised by Prof. Eric Bittner

Houston, TX

- **Thesis:** *Harnessing Entanglement and Noise in Quantum Communication Channels*
- **Anticipated graduation:** 12/2026
- **Committee:** Prof. Eric Bittner, Prof. Pavan Hosur, Prof. Claudia Ratti, Prof. Gemunu Gunaratne, Dr. Yu Zhang

National Institute of Technology Tiruchirappalli

July 2017 – May 2021

B.Tech with a Major in Mechanical Engineering

Trichy, India

Publications/Preprints

- *Programmable Dissipation via Partial Quantum Error Correction*
S. Dambal, Michael AD Taylor, and Yu Zhang
[arXiv:2605.30217](#) [quant-ph]
- *Thermalization Regimes in a Chaotic Tavis-Cummings Model*
S. Dambal and E. R. Bittner
J. Chem. Phys. 165, 074103 (2026)
- *Noise-Induced Thermalization in Quantum Systems*
S. Dambal, Y. Zhang, E. R. Bittner, and P. Hosur
[arXiv:2512.14842](#) [quant-ph]
- *Harnessing Intrinsic Noise for Quantum Simulation of Open Quantum Systems*
S. Dambal, A. Sone, and Y. Zhang
[arXiv:2510.21075](#) [quant-ph]
- *Scattering Theory of Frequency-Entangled Biphoton States Facilitated by Cavity Polaritons*
A. Piryatinski, N. Jacobus, **S. Dambal**, E. R. Bittner, Y. Zhang, and A. R. S. Kandada
Phys. Rev. A 113, 033730 (2026)
- *Quantum Spectroscopy with Biphotons: Lyapunov-based Input-Output Dynamics*
S. Dambal, A. R. S. Kandada, and E. R. Bittner
Phys. Rev. A 112, 053715 (2025).

External Research Visits

Microsoft Quantum

June-August '26

Quantum Applications Intern

Redmond, WA

- Collaborated with the Quantum Applications and Quantum Error Correction teams within the Microsoft Quantum organization.
- Devised novel methods to extend the organization's effort of simulating model Hamiltonians to open quantum systems (*Manuscript in preparation*).

Los Alamos National Laboratory

May-August '25

Graduate Research Assistant at the mQED Lab / T-1 Theoretical Division

Los Alamos, NM

- **Result:** Submitted the manuscripts:
 - * "Harnessing Intrinsic Noise for Quantum Simulation of Open Quantum Systems"
 - * "Noise-Induced Thermalization in Quantum Systems"
 - * "Scattering Theory of Frequency-Entangled Biphoton States Facilitated by Cavity Polariton"(Now published in *Phys. Rev. A*).

June-August '24

- **Result:** Studied the Optical and Transport Properties of Hyperbolic Phonon Polaritons (*Manuscript in preparation*).

Technical Skills, Open Source Contributions, and Certifications

- **Open Source:** Expanded the capability of a module to display the degree of rotation in unitary gates in Qiskit.
- **Languages/Frameworks:** C/C++, Python, Qiskit, Q#, QDK/Chemistry, Git, QuTiP, Mathematica
- **Certifications:** [Neural Networks and Deep Learning \(Coursera\)](#), [IBM Quantum Challenge: Spring 2023](#)

Academic Service

- **Referee:** Physical Review Letters, Physical Review Research, Physical Review A, Quantum Science and Technology
- **Co-Review:** Journal of Chemical Physics, Quantum Science and Technology, ACS Omega, IEEE Access

Research Communication

- **Poster:** *"Leveraging intrinsic noise to simulate open quantum systems"*, [Princeton Quantum Technology Conference 2026](#)
- **Invited Talk (30 mins):** *"Quantum Computers are inherently open quantum system simulators:"* Quantum Information Science team at Microsoft Quantum
- **Poster:** *"Thermalization Regimes in a Chaotic Tavis-Cummings Model"*, Physics Research Day, University of Houston
- **Poster:** *"Quantum Spectroscopy with Biphotons: Lyapunov-Based Input-Output Dynamics"*, [Texas Quantum Winter School, Woodcreek TX \(Jan., 2026\)](#)
- **Invited Talk (40 mins):** *"Quantum Spectroscopy with Biphotons: Lyapunov-Based Input-Output Dynamics"*, [Telluride Science and Research Center, Telluride CO \(Sept., 2025\)](#).
- **Talk (10 mins):** *"Harnessing Noise for the Simulation of Open Quantum Systems (10 mins)"*, Theoretical Division Lightning Talks, Los Alamos National Laboratory, Los Alamos NM (July, 2025)
- **Poster:** *"Quantum Spectroscopy with Biphotons: Lyapunov-Based Input-Output Dynamics"*, [Excited State Processes, Santa Fe NM \(June, 2025\)](#).
- **Talk (10 mins):** *"Quantum Spectroscopy with Biphotons: Lyapunov-Based Input-Output Dynamics"*, APS Global Summit, Anaheim CA, (March, 2025)
- **Talk (10 mins):** *"Quantum Spectroscopy with Biphotons: Lyapunov-Based Input-Output Dynamics" (10 mins)*, Texas Section APS, Houston TX (March, 2025)
- **Talks (3 mins):** *"Why can't noise be a good thing?"*, 3-minute thesis competition, among all graduate schools in UH (*finalist*), and Physics Department at UH (*winner*), Houston TX (Nov., 2024)

Academic Scholarships and Awards

- **NSM Alumni Association** (Eckhard Pfeifer/Carolyn Farb) **Scholarship** (Fall '25 and Spring '26)
- **Cullen Travel Grant:** Received \$750 to support my travel expenses and present my work at the APS Global Summit, Anaheim, CA (March, 2025)
- Finalist at the graduate-level 3-minute thesis competition among 16 graduate schools and received \$250 cash award.
- Winner of the University of Houston Physics Department 3-minute thesis competition and received \$50. (Oct. 2024)
- Recognized as **Partner at Qiskit Fall Fest lead** by IBM to organize the [IBM Qiskit Fall Fest at the University of Houston](#) among a 100 applicants world-wide. (2023)
- **Scholarship for UG: \$6000 p.a. (80% tuition)** conferred by DASA, MHRD, Govt. of India (2017-2021)

Teaching

Undergraduate Physics Laboratories

September 2022 - May 2023

- **Electricity, Magnetism, and Optics:** Taught an average class of 22 undergraduate students from non-engineering majors for 2 semesters.
- **Mechanics:** Taught 20 undergraduate engineering students for 2 semesters on carrying out 12 experiments in Mechanics.